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PAPER_1101: SCm Qubit T$_2$ Coherence via $F_{U,Bi}/F_U$ Ratio and Thermal Protection

Abstract

We derive the Star Cradle Mechanics (SCm) qubit coherence time T_2 formula, which incorporates the buoyancy field ratio $F_{U,Bi}/F_U$ and an exponential thermal protection factor from the SCm gap energy Δ_{SCm}:

$$T_2 = \frac{\hbar}{\Delta_{\mathrm{SCm}}} \cdot \exp\left(\frac{\Delta_{\mathrm{SCm}}}{k_B T}\right) \cdot S_{26}^{(3)} \cdot \frac{F_{U,Bi}}{F_U}$$

This extends conventional qubit T_2 models by encoding the UQFF buoyancy protection mechanism and the 26-dimensional phonon coupling.

1. Introduction

In superconducting and topological qubit architectures, the transverse relaxation time T_2 governs the duration over which quantum phase coherence is maintained. Standard models relate T_2 to the energy gap Δ and temperature T via:

$$T_2^{\mathrm{std}} = \frac{\hbar}{\Delta} \cdot \exp\left(\frac{\Delta}{k_B T}\right)$$

The SCm framework introduces two additional factors:

1. The **26-dimensional buoyancy prefactor** $S_{26}^{(3)}([\mathrm{SSq}]) = (1 - [\mathrm{SSq}])^3$
2. The **buoyancy field ratio** $F_{U,Bi}/F_U$ capturing the fraction of the unified field contributing to topological protection

2. SCm Gap Energy

The SCm gap energy is derived from the characteristic 1.25 THz phonon mode:

$$\Delta_{\mathrm{SCm}} = \hbar\omega_{\mathrm{phonon}} \cdot 2\pi = \hbar \cdot 2\pi \cdot 1.25 \times 10^{12}\ \mathrm{Hz}$$

$$\Delta_{\mathrm{SCm}} \approx 8.27 \times 10^{-22}\ \mathrm{J} \approx 5.17\ \mathrm{meV}$$

3. The SCm T_2 Formula

$$T_2^{\mathrm{SCm}} = \underbrace{\frac{\hbar}{\Delta_{\mathrm{SCm}}}}_{\text{base time}} \cdot \underbrace{\exp\left(\frac{\Delta_{\mathrm{SCm}}}{k_B T}\right)}_{\text{thermal suppression}} \cdot \underbrace{S_{26}^{(3)}}_{\text{26D buoyancy}} \cdot \underbrace{\frac{F_{U,Bi}}{F_U}}_{\text{buoyancy ratio}}$$

Parameter Regimes

Regime	T (K)	$\exp(\Delta/k_BT)$	Application
Dilution fridge	0.015	$\sim 10^{174}$	Superconducting qubits
^{3}He cryostat	0.3	$\sim 10^8$	Topological qubits
Liquid He	4.2	~ 14.4	High-T_c devices

Enhancement Factor

The SCm enhancement over standard T_2:

$$\eta_{\text{SCm}} = S_{26}^{(3)} \cdot \frac{F_{U,Bi}}{F_U}$$

For [SSq] = 0.57 and $F_{U,Bi}/F_U = 0.3$:

$$\eta_{\text{SCm}} = 0.0795 \times 0.3 = 0.0239$$

This represents a buoyancy-mediated coherence channel distinct from conventional T_2 decay pathways.

4. Physical Interpretation

The $F_{U,Bi}/F_U$ ratio encodes the fraction of the unified buoyancy field available for topological qubit protection. When the buoyancy component dominates ($F_{U,Bi}/F_U \to 1$), the qubit is maximally protected by the phonon-buoyancy shield. When $F_{U,Bi} \ll F_U$, other force components (magnetic, gravitational) dominate and coherence protection diminishes.

5. Implementation

Calculator: `SCmQubitT2CoherenceFUBiRatioCalculator` in CondensedPhysics.py

- Accepts temperature T, field strengths F_U and $F_{U,Bi}$, phonon frequency, and [SSq]
- Outputs T_2^{SCm}, T_2^{std}, enhancement factor, and thermal exponential

6. Conclusion

The SCm T_2 formula provides a first-principles connection between UQFF buoyancy physics and qubit decoherence, predicting coherence modulation through the $F_{U,Bi}/F_U$ ratio and the 26-dimensional phonon coupling.

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